

Editorial Notes — "Metabolic Modelling of Plant-Microbe and Microbe-Microbe Interactions in Barley Rhizosphere Synthetic Communities"

Final-editor pass over the full 69-page PDF. Issues are grouped by section/page and roughly ordered by severity within each group. Page numbers refer to the printed page numbers at the bottom of each page (not PDF page index).

⚠ Highest priority — possible factual/data inconsistency

p.6 vs p.19–20 (member counts don't match): Background section states "*Core strains were assembled to the core SynCom, HvSC1, **its 29 members** reflect taxonomic composition...*" and "*trait-prioritised SynCom, HvSC2, was assembled based on **26 strains**...*" But the Results section says "*The resulting core community contains **27 members**...*" and drop-out experiments reference "***27** (HvSC1) and 26 (HvSC2) drop-out communities.*" **HvSC1 is given as both 29 and 27 members — this needs to be reconciled**, since it affects the numeric claims throughout Results (e.g. "5.32% of all community interactions" for strain 892, drop-out counts, etc.).

Introduction (p.1–4)

p.1: "*the plant microbiome is made up of communities of microorganisms that inhabit the **internal and external of plants***" — missing noun; needs e.g. "internal and external **surfaces/tissues** of plants."

p.5: "*Rather than acting as completely isolated domains, **the plant's two main spatial regions: the above-ground phyllosphere and the below-ground rhizosphere.***" — sentence fragment, no main verb. Needs rewriting, e.g. "...the plant has two main spatial regions: the phyllosphere and the rhizosphere, which interact dynamically."

p.5: "*The rhizosphere is the zone around plant roots, **this zone is heavily enriched**...*" — comma splice; use a period or semicolon.

p.5: "*the nutritional value of **root released metabolites***" — should be hyphenated: "root-released metabolites."

p.6: "*Core strains were assembled to the core SynCom, HvSC1, **its 29 members reflect** taxonomic composition...*" — comma splice/run-on. Should be "...HvSC1, **whose** 29 members reflect..." (also see the member-count flag above).

Background — Section 2.2 (p.9–13)

p.9: "(i) Use databases...(ii) Generate draft model. **(il)** Curate the draft model." — typo, should be "(iii)".

p.9: heading "**Biomass in metabolic models**" is printed twice in a row (duplicate heading) before the paragraph starts.

p.9: "the objective function $Z = cTv$... obtain a flux distribution v that **optimised** the objective function" — tense inconsistent with surrounding present tense; should read "optimises."

p.9: "...then dividing the overall sum of all the products by 1000 **Lieven et al., 2020.**" — citation is missing parentheses: "(Lieven et al., 2020)."

p.9: "...which yields h^{-1} **Machado et al., 2018.**" — same issue: "(Machado et al., 2018)."

p.9: "thus **minimizing** the overall enzymatic..." — spelling inconsistency: the thesis otherwise consistently uses British spelling (minimising, optimisation, utilise). "Minimizing" here is the only American spelling I found; should be "minimising" for consistency.

p.10 (Fig. 2.2 caption): "each **reactions** has an associated flux value" — subject-verb agreement; should be "each reaction has."

p.10 (Fig. 2.2 caption): "Solving the equations **required** the definition of an objective function..." — tense shift mid-caption (surrounding sentences are present tense); consider "requires."

p.12: "identifying **duplicated**, dead ends, and thermodynamically infeasible loops" — "duplicated" needs a noun, e.g. "duplicated reactions, dead ends, and..."

Methods (p.16–18)

p.16: *"...the lower bounds got constrained to -1000 mmol/(gDw h). **This** tiered medium..." — missing space after the period.

p.16: unit inconsistency: "gDw" here vs. "gDW" used elsewhere (e.g. p.9 " $\text{mmol} \cdot \text{gDW}^{-1} \cdot \text{h}^{-1}$ "). Pick one consistently.

p.17 (Table 3.1): category "Amino Acids, Peptides & **Polyamine**" — should likely be plural "Polyamines" to match the other category names.

p.18: "...27 (*HvSC1*) and 26 (*HvSC2*) drop-out communities were built, **by omitting**..." — comma before "by omitting" is unnecessary; remove it.

Results (p.19–31)

p.19: "The basis for all my analysis **form** the reconstructed metabolic models" — subject-verb agreement; either "analysis **forms**" or "analyses form."

p.19: "The latter **explains also** why there are more mass imbalances..." — awkward word order; "also explains why" reads better.

p.19: "I achieved **stoichiometric consistent** models" — needs the adverb: "stoichiometrically consistent."

p.20: "the consistency score improved **35%**" — missing "by": "improved **by** 35%" (parallel to "increased by 13%" earlier in the same sentence).

p.21 (§4.2 heading text): fine, no issues noted.

p.24: "Community interactions account for almost half of the total uptake fluxes in the trait-optimised community, whereas **the more than 60%** of the core communities' uptakes..." — extra "the"; should read "whereas more than 60% of the core **community's** uptakes..." (also singular possessive: it's one core community).

p.25 (Fig. 4.2 caption): "While HvSC1 **is generally exchanges** more metabolites..." — grammatical error; should be "HvSC1 generally exchanges more metabolites" (delete "is").

p.25: "Overall, HvSC1 exchanges more metabolites distributed across smaller fluxes and **wider spanned** across the community." — awkward phrase; consider "more widely spread across the community."

p.25: "HvSC2 features **less** exchanged metabolites but a substantially **higher flux volumes** per interaction." — "less" → "fewer" (countable noun); "a...volumes" is a number-agreement mismatch — either "a...higher flux volume" or drop "a."

p.28: "...the trait-optimised community, **which's** growth rate fluctuates notably..." — "which's" is not standard English; should be "**whose** growth rate fluctuates."

p.28: "This raises the question **how** the removal of individual strains affects..." — missing "of": "the question **of how**..."

p.28: "In particular strains 1174 and 1234..." — missing comma: "In particular, strains 1174 and 1234..."

p.30: "Median individual growth rate did not change, **rather** the values spread more widely..." — comma splice; use a semicolon or period: "...did not change; rather, the values spread..."

p.30 (Fig. 4.5 caption): "**HvSC2+ Drop-In** 892 shows higher community growth rate $\mu_{2,drop-in}$ compared to μ_2 ..." — missing space ("HvSC2 + Drop-In"); also the comparison symbol " μ_2 " doesn't match the " $\mu_{2,drop-in}$ " notation used a few words earlier — check whether it should be μ_{c2} .

Discussion (p.32-43)

p.32: "...in planta /**in vitro** contextualises..." — inconsistent spacing around the slash: "in planta**/**in vitro" or "in planta / in vitro," pick one style.

p.32: "...production and secretion of antibiotics, bacteriocins, **toxins** (García-Bayona..." — missing "and" before the last list item: "antibiotics, bacteriocins, **and** toxins."

p.32: "Databases lack information **of** reactions associated with secondary metabolism" — preposition; should be "information **on/about** reactions."

p.32: "**Another challenge poses** the representation of antagonistically acting metabolites in GEMs" — inverted/awkward syntax; should read "**The representation of antagonistically acting metabolites in GEMs poses another challenge.**"

p.33: "strain 892*'s metabolic profile" — wrong apostrophe character (opening curly quote ' instead of '); should be 892*'**s.

p.33: "Members of the genus *Bosea* are classified as chemolithoheterotrophic, which **reflects in** strain 892's metabolic profile." — "reflects in" is not idiomatic; should be "which **is reflected in.**"

p.34: "strain 892 is a persistent member of the core community over a **six days time period**" — should be "**six-day** time period" or "**a period of six days.**"

p.34: "As the authors did not investigate detailed mechanisms of microbial cooperation or **conducted** community **wide** drop-out experiments" — verb should stay parallel with "did not... conduct" (not "conducted"); also "community wide" should be hyphenated "community-wide."

p.36: "Further **its** more balanced individual growth rates..." — missing comma after the transition word: "Further, its more balanced..."

p.36: "...explaining why HvSC2 shows a more unevenly distributed exchange network **among** larger asymmetries..." — "among" reads oddly here; consider "**alongside**" or "**with.**"

p.36: "...scalability caused by narrow host and environment adaptability, **i.e a community...**" — "i.e." is missing its second period/comma: "i.e., a community..."

p.36: "The selected traits for HvSC2 by Mahdi et al., 2026 focused on plant-growth promoting or pathogen-protecting traits, community performance and stability of community-host associations could be improved by additionally screening..." — this sentence appears to be missing a connective (e.g. "**While** the selected traits... focused on X, community performance... could be improved by..."). As written it reads as a run-on/fragment and needs restructuring.

p.37: "if a **strains's** function is redundant..." — extra "s": should be "strain's."

p.37: "Combining both approaches shows, **that** the predicted stabilities..." — comma before "that" is incorrect; remove it: "shows that the predicted stabilities..."

p.38: "One example, directly influencing growth rates, **are** sparsely characterised transport reactions" — subject-verb agreement; singular "example" needs "**is.**"

p.39: "which shows **substantial** metabolic shift" — missing article: "a substantial metabolic shift."

p.40: "...similarly to MICOM's original **cooperative_tradeoff()function...**" — missing space before "function."

p.40: "One example **are** non-growing or dormant strains..." — same agreement issue as above; "example **is.**"

p.41: "As for the drop-in community **no detailed analysis...**" — missing comma: "As for the drop-in community, no detailed analysis..."

p.43 (opening of Conclusion, p.44 technically): "A core community (HvSC1), assembled from co-occurring strains, and a trait-optimised community (HvSC2), composed of strains previously shown to promote plant growth or protection." — this reads as a sentence fragment (no main verb); either merge with the previous sentence or add a verb.

p.44: "The thesis also discusses limitations and opportunities to overcome these in both from the

p.44: "The thesis also discusses limitations and opportunities to overcome these **in both from** the computational and the biological perspective." — extra "in"; should be "**from both** the computational and the biological perspective."

p.42: "Accuracy would be improved by removing EGCs, customising biomass functions, **besides** including more experimental data..." — "besides" doesn't fit the list structure; consider "as well as including..." or restructure as a proper list.

p.42: "even with remaining uncertainties **e.g.** in model reconstruction" — missing comma after "e.g.": "e.g., in model reconstruction."

p.42: "quantifying the cooperation between community members **i.e via** systematic pairwise co-culture assays" — "i.e." needs punctuation: "i.e., via."

p.42: "Analysing differences in metabolic requirements, community interactions and perturbation robustness, **provides**..." — the comma before "provides" incorrectly separates subject from verb; remove it.

p.43: "combining core with trait-selected strains, **improves** community stability..." — same issue; remove the comma before "improves."

Recurring style points (not necessarily "errors," but worth a consistency pass)

"can not" vs "cannot": the thesis uses "can not" as two words in several places (p.10, p.35, p.31) but "cannot" as one word elsewhere (p.35, p.40). Pick one consistently — "cannot" is the more conventional academic form.

British vs. American spelling: overwhelmingly British ("minimising," "optimisation," "colour," "characterised") with a couple of American slips ("minimizing" on p.9, "Optimizer" in the Table 2.1 citation — the latter may be an unavoidable proper-noun/product name, so probably fine).

Bibliography capitalisation: several reference titles have species/genus names in lowercase that should be capitalised — e.g. "vibrio natriegens" (Lee et al., 2019), "escherichia coli" (Mahadevan et al., 2002; Schuetz et al., 2007), "bacterium bosea sp." (Zhang et al., 2019). This is likely a BibTeX/LaTeX capitalisation artefact (missing `{ }` protection around the species name) rather than something typed by hand, but it's worth a global fix in the .bib file.

Soare et al., 2023 reference (p.50): "Dobre, C.-A., & **Firăt, oiu, A. R.**" — the author name looks garbled/mis-parsed (likely a diacritic encoding issue in the .bib entry, probably "Firăţoiu, A. R."). Worth checking the original source and fixing the .bib entry.

Note: This review focused on grammar, punctuation, typos, and internal consistency. I did not verify scientific claims, citation accuracy, or statistical correctness — only the member-count discrepancy above stood out as a potential factual issue worth double-checking against your data.